NEBIUS × NVIDIA GLOBAL AI HACKATHON · BEST APPS AND AGENTS

Storefront Fraud Investigator

An agent that answers a fraud analyst’s questions about an online store in plain English, with the numbers it rests on.

NVIDIA Nemotron on Nebius Token Factory · try it at https://zfcrow.online

THE PROBLEM

Fraud rarely sits in one row

One account takeover, as the store records it: four signals from the same session, in three different tables. Each looks harmless alone. Together they’re the case.

sessions
First-time device and country

+

events
Password reset

+

events
Delivery address changed

+

orders
Order 100× the customer’s usual

Nothing is pre-flagged. An analyst joins these by hand. The agent does the joins, compares against baselines and shows its reasoning — every number traced to a query.

NEBIUS × NVIDIA GLOBAL AI HACKATHON · BEST APPS AND AGENTS

PROJECT DEMO

NVIDIA Nemotron on Nebius Token Factory · zfcrow.online

HOW IT WORKS

Every call is Nemotron on Token Factory

Reason

Nemotron 3 Super decides whether more data is needed and which tool gets it.

AGENT tier
Nemotron 3 Super

→

Tools, up to 3 calls

- SQL on Postgres, read-only role
- Session search: Qwen3 embeddings + BM25
- Tavily: is this a known scheme?

AGENT tier · writes the args
Nemotron 3 Super

→

Answer

Every number from a query result, ending with a reasoning trace.

Tier picked per run
Fallback: AGENT

→

Judge

Nemotron scores faithfulness, coverage and tool routing.

MODEL_JUDGE
Nemotron 3 Super

The answer's tier is picked per run by synthesis_tier — AGENT is only the fallback

2+ tool results → REASONING
Nemotron 3 Ultra

1 result → AGENT
Nemotron 3 Super

Out of scope → FAST
Nemotron 3.5 Lightning

RESULTS · ONE REAL RUN

How one question was routed

Analyst: *“Was there anything unusual about Samuel Hale’s most recent order?”*

1 · REASON → query_sql
Super · 3.2 s

“The first step is to identify Samuel Hale’s customer record...”

SQL (schema read first)
$4,246 order to Bucharest, new device, Romanian IP

→

2 · REASON → query_sql
Super · 1.6 s

“...compare this order to his historical behavior (earlier orders/sessions)”

SQL: baseline without this order
2 earlier orders, avg $38, one device, Singapore only

→

3 · REASON → query_sql
Super · 2.6 s

“...need the historical ship country to see if Romania is new.”

SQL: his 2 earlier orders
$22 and $54, mobile, Singapore

→

4 · WRITE THE ANSWER
Ultra · 5.2 s

All 3 calls used, so it answers. 3 results → Ultra writes it; Lightning adds follow-ups.

Answer
Account takeover: order 112× his usual. Hold order 196.

Enforced in code: schema read before every query · web search only after store evidence · max 3 tool calls · rejected SQL repaired free

Routing vs all-Ultra: 8 LLM calls to answer: 6 on Super, 1 on Ultra (the answer), 1 on Lightning (follow-ups). All-Ultra = 8 Ultra calls.
One recorded run. Paths vary between runs: all SQL, or SQL + session search. Answer checked against the planted incident; all data is fictional.

try it at https://zfcrow.online · Code: github.com/ZFCrow/fraud-investigator · MIT

NEBIUS × NVIDIA GLOBAL AI HACKATHON · BEST APPS AND AGENTS

THANK YOU!

try it at https://zfcrow.online · Code: github.com/ZFCrow/fraud-investigator · MIT